

FlowRecon-Bench: A benchmark for sparse-sensor flow-field reconstruction across flow regimes, with calibrated uncertainty

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Reconstructing a full flow field from sparse point sensors is a core task of experimental and operational fluid mechanics, now addressed by a growing machine-learning literature of deterministic operator learners and conditional generative models. The field has no shared evaluation: published methods use incompatible, often temporally leaky splits, ensembles are drawn but not scored with proper probabilistic metrics, and classical baselines are rarely reported. We present FlowRecon-Bench, a benchmark spanning three flow regimes of increasing dimensionality and dynamical complexity — laminar cylinder vortex shedding across Reynolds numbers, two-dimensional Kolmogorov turbulence, and three-dimensional isotropic turbulence (JHTDB) — with leakage-audited splits, frozen sensor manifests, per-channel accuracy, proper scoring rules (CRPS, coverage, spread-error), physics metrics, and matched cost accounting. Across eleven learned methods and four classical floors, method ranking is regime-dependent and no method wins twice: gappy POD with 20 modes wins the laminar cylinder outright (0.056 relative L_2 , including the never-observed pressure), a deterministic operator learner leads two-dimensional turbulence (Geo-FNO, 0.417) yet is beaten on CRPS by three of four generative models, and latent flow matching, a point-token transformer, and plain interpolation split the channels of 3D turbulence. No method recovers the unobserved channels of 3D turbulence; nearly every ensemble is under-dispersed, and split-conformal recalibration restores stated coverage. Benchmark scores are systematically worse than literature numbers on the same data, and controlled ablations attribute the gap to evaluation protocol rather than model capability: the same checkpoint improves by 25% when scattered sensors are rearranged onto a uniform lattice, whereas re-scoring frozen checkpoints on their own training frames changes their observed-channel error by at most 5% in both 2D and 3D, so split leakage is a correctness requirement rather than the source of the gap. Datasets, splits, sensor manifests, configurations, and the evaluation driver are released as a versioned artifact. [\[TODO: artifact DOI\]](#)

I. INTRODUCTION

Flow fields govern systems that experiments can only observe pointwise: particle-image velocimetry resolves planes of a turbulent flow while the surrounding volume goes unmeasured, flight-test campaigns record surface pressure at taps while the off-body flow that determines performance is inaccessible, and environmental monitoring networks sample the atmosphere at stations kilometers apart. Recovering the full field from such sparse, indirect data is a long-standing inverse problem,^{1–3} and at realistic sensor densities it is severely ill-posed: many fields fit the data, and the missing information sits disproportionately at the small scales. A machine-learning literature now more than a decade deep addresses it with deterministic reconstruction networks,^{4–7} and, more recently, with conditional generative models that sample the distribution of fields consistent with the observations rather than regressing to its mean.^{8–13}

What that literature does not have is a shared evaluation. Community benchmarks for scientific machine learning — PDEBench,¹⁴ The Well,¹⁵ CFDBench,¹⁶ AirfRANS,¹⁷ and their successors — pose forward-surrogate or uniform-downsampling super-resolution tasks,

not reconstruction from scattered point sensors; the Johns Hopkins Turbulence Database is a query service with no canonical protocol, and published sparse-reconstruction papers each cut their own incompatible extraction from it. Three consequences follow, and this paper documents all three. First, *splits leak*: turbulence snapshots are strongly autocorrelated in time (frame-to-frame correlation 1.00 at unit separation on our 3D data), so the standard random-in-time split makes every test frame a near-duplicate of a training frame. We expected this to dominate and measured it directly: re-scoring frozen checkpoints on their own training frames — the strongest leak a split can produce — changes their observed-channel error by at most 5% in both 2D and 3D, and the one larger gain sits on an unobserved 3D channel that no split makes recoverable (Sec. V G). At benchmark data scales these models learn the sensing operator rather than a lookup table, so leakage control is a correctness requirement, not the explanation for the gap to the literature. Second, *ensembles are drawn but not scored*: generative reconstruction papers routinely show mean-and-standard-deviation maps, several claim calibrated uncertainty on that evidence alone, and, to our knowledge, none reports a proper scoring rule or a calibration diagnostic for 3D single-snapshot reconstruction from scattered sensors;^{18–22} every scored precedent in adjacent settings is under-dispersed. Third, *classical baselines are absent*: training-free interpolation and gappy proper-orthogonal-decomposition floors are rarely reported, and we find that inverse-distance weighting matches or beats all but one deep model’s observed-channel error on 3D turbulence at 1% sensor density, and that gappy POD wins the laminar regime outright. A consequence worth stating plainly up front: every number and image in this paper will look worse than its counterparts in the literature, and Sec. V G shows by controlled ablation — same checkpoints, protocols swapped — that this gap is the protocol, not the models.

FlowRecon-Bench is our response: a benchmark built to answer the question a practitioner actually has — *which method family should I use, in which flow regime, at what sensor budget, with what uncertainty and at what cost?* It spans three regimes forming a ladder in dimensionality and dynamical complexity (Sec. II B): laminar cylinder vortex shedding, where classical linear methods should be near-optimal; two-dimensional Kolmogorov turbulence, the standard testbed of the 2D generative literature;^{8,9} and three-dimensional isotropic turbulence under a spatially disjoint protocol in which sensors observe only two of four channels. Every method is evaluated under identical seeded sensor draws, leakage-controlled splits with a strict tune/test separation, per-channel accuracy, proper scores and calibration diagnostics, windowed spectral metrics, and measured training and inference cost.

Our contributions are the benchmark and the findings it forces. (i) A three-regime task suite with leakage-audited splits, frozen sensor manifests, and a fingerprinted evaluation driver, released as a versioned artifact together with per-method configurations and audit notes. (ii) A fidelity-audited fleet of eleven learned methods and four classical floors spanning deterministic operator learners and latent and ambient generative routes, parameter-matched where the published configurations allow and budget-disclosed where they do not (Sec. III). (iii) Findings that generalize beyond any single table: method ranking is regime-dependent; no method recovers channels the sensors do not observe in 3D turbulence (a fleet-wide identifiability limit, not a per-method defect); accuracy and sample realism trade off through the sampler’s operating point, so ranking generative models on mean-field error alone is misleading (Sec. V A); nearly every scored ensemble is under-dispersed, and a post-hoc recalibration fitted on a disjoint tuning split repairs stated coverage (Sec. V D); and the computational axis separates grid methods, whose memory walls we measure, from point methods, whose costs are flat in resolution (Sec. V F).

TABLE I. The three regimes of FlowRecon-Bench. Each step of the ladder adds one source of difficulty; the split column names the leakage-controlled protocol (Sec. II C).

Regime	Dataset	Points	Channels	Split	Axis isolated
2D laminar periodic	Cylinder wake (new)	2.4×10^4	u, v, p	cross-Re	low intrinsic dim.
2D chaotic	Kolmogorov 256 ²⁸	6.6×10^4	ω	cross-traj.	2D multiscale
3D chaotic	JHTDB isotropic ²³	1.95×10^6	U_x, U_y, U_z, p	spatial	unobs. channels

II. BENCHMARK DESIGN

A. Task

Let $u : \Omega \rightarrow \mathbb{R}^C$ be a flow field drawn from a distribution over fields (snapshots of a turbulent flow, solutions over varying geometry). It is observed through a measurement operator returning readings $\mathcal{O} = \{(\mathbf{x}_j, c_j, y_j)\}_{j=1}^M$ with $y_j = u_{c_j}(\mathbf{x}_j) + \varepsilon_j$, where $\mathbf{x}_j$ is a sensor location, c_j indexes the observed quantity, and ε_j is noise. The task is single-snapshot reconstruction: estimate u — or, for probabilistic methods, sample an approximation to $p(u \mid \mathcal{O})$ — at all query locations, with no access to sensor time histories, initial conditions, or the governing solver. This covers sensors scattered arbitrarily or confined to a surface, observed quantities forming a subset of the target channels, and corrupted readings. Sequential assimilation and forward surrogacy are deliberately out of scope; they are different problems with different information budgets.

B. Datasets: three flow regimes

The three regimes form a ladder in both spatial dimension and dynamical complexity — 2D laminar and periodic, 2D chaotic, 3D chaotic — so that each step isolates one source of difficulty while holding the task fixed (Table I); preprocessing details and download/regeneration recipes ship with the artifact.

a. Cylinder wake (new). Two-dimensional incompressible flow past a circular cylinder, generated with OpenFOAM 9 (`pimpleFoam`, laminar) on a 12-block O-grid of 23,800 cells with boundary-layer grading, at $\text{Re} \in \{60, 80, 100, 150, 200, 250\}$ ($D = U_\infty = 1$). Each case runs a transient phase to $t = 120$ and a sampling phase to $t = 270$ written every 0.5 time units: 300 snapshots, roughly 25 shedding cycles at $\text{Re} = 100$. The physics-level validation is the Strouhal number from the lift spectrum, $\text{St} = 0.140, 0.160, 0.173, 0.187, 0.200, 0.207$ at $\text{Re} = 60\text{--}250$ against the empirical 0.135, 0.164, 0.183 and 0.196 at $\text{Re} = 60, 100, 150$ and 200;^{24,25} the 3–5% excess is the familiar two-dimensional over-prediction, and the trend and spacing are reproduced. Mean drag lies between 1.40 and 1.46, consistent with the 6% blockage, and lift amplitude grows monotonically with Re . The dataset ships as the native cell-center point cloud, with the wall-adjacent cell ring indexed for the surface-tap variant, and as a 400×200 uniform-grid interpolation over $x \in [-3, 17]$, $y \in [-5, 5]$ with a body mask for grid-locked methods. Cross-Re protocol: train on $\text{Re} \in \{60, 100, 150, 200\}$, evaluate on the held-out $\text{Re} \in \{80, 250\}$ (one interpolative, one mildly extrapolative), with temporally blocked splits inside the training cases. An observe- u -only variant probes cross-channel inference (v, p unobserved) in a regime where the shedding phase makes the unobserved channels largely determined by the observed one.

b. Kolmogorov turbulence. The two-dimensional forced-turbulence vorticity dataset of Ref. 8, standard in the 2D generative-reconstruction literature: Kolmogorov flow on the doubly periodic box $[0, 2\pi)^2$ at Reynolds number 1000 with wavenumber-4 forcing, 40 independent trajectories of 320 frames at 256^2 . We stride time by 4 (80 frames per trajectory, 3,200 total) and hold out entire trajectories: 32 for training, 8 for evaluation, assigned by a fixed seeded permutation. Cross-trajectory holdout is the 2D analogue of the spatial holdout below — reconstruction is scored on realizations of the flow the model has never seen

at any time.

c. 3D isotropic turbulence. Four 125^3 cutouts at disjoint spatial locations of the JHTDB `isotropic1024coarse` DNS²³ (forced isotropic turbulence at $Re_\lambda \approx 418$; each cutout spans ≈ 0.56 integral length scales at a grid spacing of ≈ 2.1 Kolmogorov scales), 50 snapshots each at a stride of 100 database steps, fields (U_x, U_y, U_z, p) , per-field standardized. Training uses cutouts 1–3; all reported evaluation uses cutout 4, spatially disjoint from everything seen in training. Sensors observe U_x and U_z only; U_y and p are never observed, making this the identifiability probe of the suite.

C. Splits and leakage control

Every split in FlowRecon-Bench is chosen against the autocorrelation structure of its data, and the protocol is enforced by tooling rather than convention. Turbulence snapshots correlate at $r = 1.00$ frame-to-frame and $r = 0.67$ at separation 100; random-in-time splits therefore risk measuring near-duplicate interpolation (Sec. V G measures how much they actually do). The suite uses spatial holdout (a disjoint cutout) for 3D turbulence, trajectory holdout for Kolmogorov, Reynolds-number holdout for the cylinder, and temporally blocked splits with decorrelation guard gaps wherever time enters. Model selection is separated from reporting: held-out snapshots are partitioned into a tuning half (odd indices) used for any checkpoint or hyperparameter selection and a test half (even indices) on which every reported number is frozen. The evaluation driver fingerprints the protocol — snapshot indices, sensor counts, and the index-sum of every seeded sensor draw — and aborts on mismatch, so no two methods can silently see different observations. The cross-checks the artifact runs automatically were added after two construction-time incidents (a baseline’s evaluation path collapsing a spatial axis, and a one-frame boundary leak from integer truncation in the split arithmetic), both caught before any number was reported.

D. Sensor protocols

The primary protocol draws sensors uniformly at random over the domain, per observed channel, at densities from 0.1% to 2% of the discretization (regime-dependent; 1% is the headline density), swept from 0.1% to 10% on Kolmogorov (Fig. 7), with counts and layouts redrawn per training step for amortized methods and frozen seeded draws at evaluation. Sensor layout is not incidental to difficulty: at fixed count, rearranging scattered sensors onto a uniform lattice changes reconstruction error by tens of percent (Sec. V G), so the layout is part of the protocol specification rather than an implementation detail. Two layout variants accompany the random-scatter default — a uniform lattice at matched count, and sensors confined to the cylinder wall ring (Sec. IV B) — and channel subsets are specified per regime (Sec. II B). Operator variants corrupt the assembled sensor sets in one place, after the draw and before any model sees them, by shared seeded code, so every method consumes bit-identical corrupted observations: Gaussian noise at $\sigma \in \{0.1, 0.3\}$ in standardized units, contiguous slab occlusion removing 25% of sensors, and channel dropout removing entire observed quantities (implemented, but not exercised in the reported results: the operator study runs on Kolmogorov, which observes a single channel; Table VIII). Corrupted runs carry their own protocol stamp so they can never be aggregated with clean ones. Distinguishing per-frame, per-trajectory, and globally fixed sensor masks matters for comparability across papers, because the three make materially different problems out of the same nominal sensor budget. FlowRecon-Bench uses per-frame draws with frozen evaluation manifests, and says so explicitly, precisely so that a disagreement with a paper using another convention can be recognized as a protocol difference rather than a result.

E. Metrics

a. Accuracy, per channel. Relative L_2 per channel is primary, reported separately for observed and unobserved channels — aggregates average two different questions and are shown flagged, never headlined. For ensemble methods both the ensemble-mean error and the single-sample error are reported; the gap between them is the smoothing an ensemble average buys, and single-sample error measures whether individual realizations are physical.

b. Probabilistic scores. For an ensemble $\{s_k\}_{k=1}^K$ and truth y at a point, the fair CRPS estimator¹⁸ is $\text{CRPS} = \frac{1}{K} \sum_k |s_k - y| - \frac{1}{2K(K-1)} \sum_{k \neq l} |s_k - s_l|$, averaged over points and fields in standardized units. The spread–error ratio is the ratio of ensemble standard deviation (RMS over points) to the RMS error of the ensemble mean (ideal 1.0); coverage₉₀ is the fraction of points whose truth falls in the central 90% ensemble interval (ideal 0.90); rank histograms accumulate the rank of the truth within the ordered ensemble. Deterministic methods enter the CRPS column exactly (CRPS reduces to MAE for a point forecast); their dispersion entries are undefined, not zero. Calibration metrics move with the operating point — sampler steps, ensemble size, sensor density — by more than they separate methods, so every calibration number carries its operating point and cross-method comparisons are made only at matched settings. Ensemble size deserves particular care: in a K -sweep on Kolmogorov ($K=8$ –64, same frames and sensors), DMF-Gen’s ensemble-mean rel- L_2 moved only 0.487→0.475 and CRPS was flat (the fair estimator is unbiased), but coverage₉₀ rose 0.59→0.73 — finite-ensemble quantiles are systematically too narrow, so $K=8$ *understates* every generative method’s calibration. We therefore report accuracy and CRPS at $K=8$ and treat $K=8$ coverage as a conservative bound, with the sweep in the artifact.

c. Physics metrics. Shell-averaged energy spectra of single samples (never ensemble means, which destroy small scales by construction), band ratios against the reference solution, structure functions, and velocity-gradient PDFs. One estimator detail is load-bearing enough to state in the protocol: non-periodic cutouts must be windowed before the FFT. On the 3D data the raw spectrum of the *truth* falls only $2.4\times$ from $k = 32$ to the Nyquist wavenumber while a variance-preserving Hann taper restores a 5.8×10^5 decay; without the taper, a smooth Gaussian-process draw with zero energy at those wavenumbers registers the same broadband floor as the DNS, and spectral ratios computed there are ratios of two leakage floors that tend to unity for every method. All spectral numbers in FlowRecon-Bench use the windowed estimator.

d. Cost. Model selection is a cost–benefit decision, so cost is a first-class metric family, measured under the same harness for every method on every dataset: training seconds per optimizer step, total GPU-hours to the finished model, peak training memory, inference wall-clock per full-field reconstruction, and peak inference memory. Rows without cost entries are treated as incomplete by the table-assembly tooling. A separate resolution-scaling study (Sec. V F) measures memory versus output discretization from 64^3 to 1024^3 , with out-of-memory walls reported as observed failures, not extrapolations.

e. Statistical resolution. Every learned row of a regime is scored on the same frames under the same seeded draws, so the per-snapshot difference between two methods is the unit of analysis. We report the mean paired difference with a 95% percentile-bootstrap interval (20,000 resamples) and call a gap *resolved* only when the interval excludes zero and a Wilcoxon signed-rank test agrees at $p < 0.05$; methods whose gaps are not resolved are reported as a tie group rather than ranked. On the two 2D regimes every method was additionally retrained under two further seeds, so a gap can be compared with the training-seed spread as well as with the frame-resampling interval.

F. Fairness methodology

Benchmark tables are only as credible as their weakest baseline, so the fleet is constructed against our own biases. (i) *Upstream fidelity audits*: every baseline implementation is diffed

TABLE II. Capability matrix: which method families can enter which regimes. RS = requires resampling to a grid, lossy near the body-fitted boundary of the cylinder mesh, the only regime here that exercises the distinction; each cylinder cell records the export the corresponding run actually opened (Senseiver, MLP-RBF and DMF-Gen read the 23,800-cell mesh; Geo-FNO, latent FM, S3GM and SiT the 400×200 grid export). [§]SiT is point-tokenized in 3D but patch-tokenized in both 2D runs: the family can enter grid-free, this configuration did not. Rows are families rather than methods because the claim is about interfaces.

Family	Cyl.	Kolm.	JHU
Interpolation/POD anchors	✓	✓	✓
Point-attention det. (Senseiver)	✓	✓	✓
Grid det. (FNO-class)	RS	✓	✓
DeepONet-class	✓	✓	✓
Latent-grid generative (LFM)	RS	✓	✓
Latent neural-field gen. (CoNFILd)	✓	✓	✓
Voxel/video diffusion (Gen4Turb, S3GM)	RS	✓	✓
Point-token generative (SiT-point)	RS [§]	✓ [§]	✓
Ambient point-cloud FM (DMF-Gen)	✓	✓	✓

against its published code before its numbers are admitted, and every declared adaptation (dimensionality ports, set encoders, sampler retuning) is documented in the per-method audit notes; audit incidents (a conditioning defect in the latent flow-matching baseline, sampler-parameter sensitivity in S3GM) were repaired or labeled before reporting. (ii) *Parameter matching*: learned methods are matched to ~ 6.5 M parameters where the published configuration allows; declared waivers (CoNFILd’s published prior at 118.9M) are printed in the table. (iii) *Identical conditions*: the same data files, splits, augmentation, normalization statistics, seeded sensor draws, and metric estimator for every row. (iv) *Budget disclosure*: optimizer steps and wall-clock are reported as run, not silently equalized, and the table notes which budgets exceed which. Auditing configured against as-run step counts for all fourteen 2D rows found thirteen within 4% of ~ 50 k steps and one budget error — Kolmogorov Geo-FNO at 200k steps, because its batch size was a quarter of the reference and the budget had been fixed in epochs; the matched rerun replaces it (Sec. IV C). (v) *Tuning-budget disclosure*: the method we co-developed (DMF-Gen) received more configuration search than any baseline; upstream-faithful configurations are frozen as the headline rows and every tuned variant is a separately labeled arm. (vi) *Selection hygiene*: checkpoint selection uses the tuning split only (Sec. II C).

III. METHODS EVALUATED

The fleet spans four families; Table II records which methods can enter which regimes, itself a benchmark result, and Fig. 1 expands it to the per-method deployment axes that Sec. V measures. No method fills every column: DMF-Gen alone clears the representation axis but is the slowest sampler; latent FM inverts that profile. Kolmogorov and the 3D cutouts are periodic Cartesian grids on which every family runs natively, so the interface filter is exercised here only by the cylinder’s body-fitted mesh, at its mildest. Per-method configurations, adaptations, and audit notes ship with the artifact.

a. Classical anchors. Four training-free floors run on every regime: nearest-neighbor interpolation (KD-tree), inverse-distance weighting over the $k = 8$ nearest sensors, gappy proper orthogonal decomposition²⁶ (POD basis from the training set, modal coefficients by least squares on the sensor readings, rank 80), and the training-mean constant. Distance handling is non-periodic by default — an earlier periodic implementation understated these floors on non-periodic cutouts and was corrected in audit — with periodic variants where

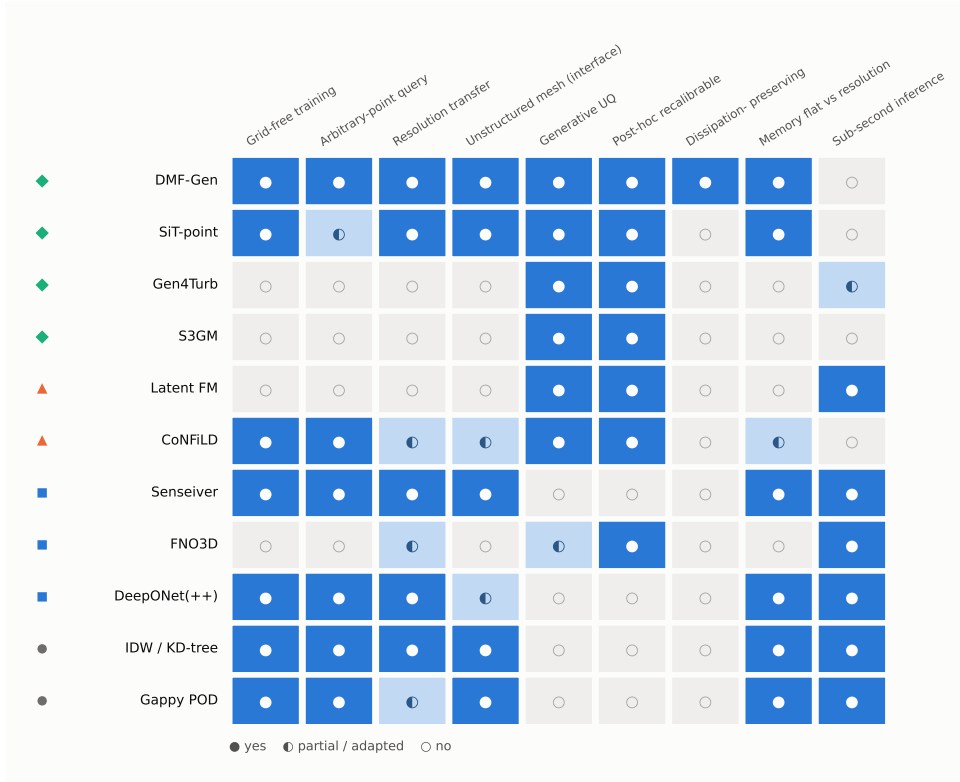


FIG. 1. Per-method capability profile across the deployment axes measured in this paper (filled = yes, half-filled = partial or adapted, open = no). Behavior under corrupted operators is a graded measurement rather than a capability and is reported in Table VIII; “dissipation-preserving” records the windowed dissipation-band ratios (Sec. V E); “memory flat vs resolution” and “sub-second inference” record the measured scaling study (Sec. V F); “recalibrable” records whether the post-hoc wrapper of Sec. V D applies. Axes are marked “no” only when the failure is structural.

the domain genuinely is periodic. As deterministic point forecasts their CRPS equals their MAE. These anchors define what any learned method must beat to justify its training cost.

b. Deterministic operator learners. Senseiver⁵ is an attention-based point-native reconstructor mapping arbitrary sensor sets to query values through a latent bottleneck; it trains under the identical sensor randomization as the amortized generative methods. Geo-FNO enters the two 2D regimes as the grid-based deterministic learner: a Fourier neural operator^{27,28} that reads the sensors rasterized onto the grid as a value channel and a binary mask channel per field and returns the full field in one forward pass, so on the cylinder it consumes the uniform-grid export. MLP-RBF is a point-native deterministic control for the same regimes: each query gathers the sensor tokens through a softmax radial-basis kernel over sensor distance plus a mean-pooled global summary and is trained by direct regression, which isolates the point-native conditioning path from the generative formulation. FNO3D places a spectral (FFT-grid) backbone inside the same conditional-generation framework as DMF-Gen, isolating the backbone axis from the formulation axis. DeepONet²⁹ enters in upstream-faithful form (branch/trunk with a declared set-encoder adaptation for variable sensor sets); its near-constant predictions are the pre-registered structural consequence of global mean pooling in the branch, and a structured-branch redesign (“DeepONet++”, multiresolution binned pooling) is reported as a labeled improvement arm. Deterministic models return conditional-mean estimates: optimal in squared error, systematically smoothed, and structurally unable to report what the sensors leave undetermined.

TABLE III. Who wins where. Leaders per regime at the headline sensor density (1%), by method family, as aggregate relative L_2 of the ensemble mean; per-channel values, CRPS and cost are in Tables IV, VII and IX. “Tie” marks a generative leader whose gap to the rows behind it is not resolved by paired intervals (Sec. II E). On 3D turbulence the aggregate averages two observed and two unobserved channels; on the observed channels alone SiT-point leads (0.143 on U_x) with interpolation second (0.174). The last column asks whether any method recovers a channel the sensors never observe.

Regime	Best classical	Best deterministic	Best generative	Lowest CRPS	Unobserved channel
Cylinder wake	Gappy POD 0.056	Geo-FNO 0.093	Latent FM 0.192	Gappy POD 0.028	recovered (p 0.051)
Kolmogorov	IDW 0.531	Geo-FNO 0.417	DMF-Gen 0.487 (tie)	S3GM 0.231	one channel only
3D turbulence	IDW 0.589	Senseiver 0.699	Latent FM 0.469	Latent FM 0.244	not recovered (best 0.60)

c. Latent generative. Latent flow matching (LFM) is the latent-diffusion-family standard:^{11,30} a convolutional autoencoder trained on full fields, followed by conditional rectified flow in its latent space with a point-encoder conditioning branch. It is the strongest aggregate performer on 3D turbulence and carries the family’s structural signature, a dissipation-range spectral floor inherited from the autoencoder.^{31,32} CoNFIL¹¹ is the published latent neural-field competitor: a conditional-neural-field autodecoder compressing each snapshot to a latent vector, an unconditional diffusion prior over latents, and diffusion posterior sampling for conditioning; three arms are reported (published prior; capacity-matched; 384-dimensional), with the latent-capacity ceiling of the autodecoder measured explicitly as the decoder bound.

d. Ambient generative. Gen4Turb¹² is voxel diffusion: an elucidated-diffusion 3D U-Net with mask-channel conditioning, run at its native 120^3 on the 3D data and retrained under the benchmark’s observation model (the most generous variant is reported). S3GM⁹ is a pretrained video-diffusion prior with sampling-time guidance, natively two-dimensional; its 3D entry is a declared adaptation treating one spatial axis as video time, and its published fixed guidance weights required tuning-split renormalization to converge at 3D scale. SiT-point adapts a scalable interpolant transformer³³ with a point tokenizer (not patchification): sensors and queries are tokenized directly, generation runs in fixed-size token chunks, and the chunk budget is the visible constraint. That is the 3D configuration; on the two 2D regimes the same implementation is run patch-tokenized, which is what those rows report, and on the cylinder it therefore reads the uniform-grid export rather than the mesh (Table II). Its single samples carry chunk-incoherent speckle that ensemble averaging hides, one reason FlowRecon-Bench reports member-level metrics. DMF-Gen³⁴ performs conditional rectified flow directly on field values at scattered points — ambient function space, no autoencoder, no voxelization — with a Gaussian-process source shared across discretizations, conditioning factored into a global cross-attention summary plus a local k -nearest-sensor retrieval per query, a Monte-Carlo subsampled training objective (a random $\sim 2\%$ of points per step), and training amortized over sensor counts, layouts, noise, and observed-channel subsets. We evaluate it as published, at the benchmark’s parameter budget; it is one row of the fleet, not the paper’s contribution.

IV. RESULTS BY REGIME

Table III collects the leader of each regime by method family; the three subsections give the evidence and the paired resolution behind each entry.

A. Laminar vortex shedding: where classical methods should suffice

The dataset was generated for this benchmark and validated against the Strouhal and drag literature before any model saw it (Sec. II B). Sensors observe (u, v) at 1% of the 23,800

TABLE IV. Cylinder wake, cross-Re evaluation on the held-out $\text{Re} \in \{80, 250\}$ (u, v observed at 1% = 238 sensors per field on the 23,800-cell mesh; p is *never observed*; learned rows: 50 Re-stratified held-out frames, $K=8$; classical floors: all 600). Deterministic rows carry $K=1$ and null dispersion. Interpolation returns the training mean on the unobserved channel, hence $p=1.000$ by construction.

Method	Rel. L_2				CRPS	Cov. ₉₀	s/field
	agg.	u_x	u_y	p			
DMF-Gen (NFE 4)	0.221	0.182	0.353	0.128	0.098	0.51	0.08
SiT (50 steps)	0.328	0.302	0.409	0.274	0.106	0.55	0.19
Latent FM (NFE 4)	0.192	0.170	0.239	0.166	0.076	0.89	0.03
S3GM (PC-DPS)	1.019	1.035	1.017	1.005	0.521	0.58	29.24
Senseiver	0.145	0.101	0.240	0.095	0.080	—	0.01
MLP-RBF	0.388	0.265	0.679	0.221	0.213	—	0.01
Geo-FNO	0.093	0.087	0.103	0.088	0.053	—	0.01
Gappy POD ($r=20$)	0.056	0.043	0.075	0.051	0.029	—	—
Gappy POD ($r=80$)	0.067	0.055	0.103	0.043	0.028	—	—
IDW ($k=8$)	0.707	0.504	0.616	1.000	0.440	—	—
Nearest neighbor	0.707	0.508	0.613	1.000	0.421	—	—
Training mean	1.000	1.000	1.000	1.000	0.664	—	—

mesh cells (238 per field); pressure is never observed; evaluation is on the held-out Reynolds numbers $\{80, 250\}$.

The classical floors decide this regime before the learned fleet reports. The training family’s POD spectrum is essentially exhausted at rank 20 (99.7% energy), and *gappy POD with 20 modes reconstructs held-out Reynolds numbers at 0.056 relative L_2 — including the never-observed pressure at 0.05* — because the modal coefficients inferred from sparse velocities carry the full coupled state. Pure interpolation cannot do this (p column: 1.0 by construction) and trails badly on the velocities as well (IDW 0.71 aggregate). This is the mirror image of 3D turbulence, where the same POD basis was useless (0.95, Table IX) and interpolation was the strong floor: *which classical method is the one to beat is itself regime-dependent*, and a learned method only earns its complexity here if it approaches the POD floor at comparable cost.

It does not, and the gap is resolved rather than marginal (Table IV): on the same 50 frames, rank-20 gappy POD beats the best learned method, Geo-FNO at 0.093, by 0.037 relative L_2 (95% CI $[-0.042, -0.032]$, $p < 10^{-14}$), and the generative models trail further (latent FM 0.192, DMF-Gen 0.221, SiT 0.328). Geo-FNO reads the grid export and receives its sensors on the grid rather than the mesh, so that pairing is over frames only; against the mesh-native rows, re-scored on the floors’ exact sensor draw (the table reports the original runs, which differ by at most 0.006), POD’s lead is resolved with the same confidence (Senseiver -0.084 , DMF-Gen -0.165 , MLP-RBF -0.339 ; all $p < 10^{-14}$). Among the learned rows DMF-Gen places fourth, behind latent FM by 0.030 ($[0.014, 0.044]$) and behind Geo-FNO by 0.128 ($[0.123, 0.134]$, better on none of the 50 snapshots). The ordering on the never-observed pressure channel is the sharper statement — gappy POD recovers p at 0.043–0.051, better than any learned method (Geo-FNO 0.088, Senseiver 0.095, DMF-Gen 0.128) — because a basis fitted to the coupled state propagates sparse velocity observations into pressure, whereas the learned models must infer that coupling from data alone. Read against Sec. V C, where *no* method recovers an unobserved channel of 3D turbulence, this is the benchmark’s cleanest statement that cross-channel identifiability is a property of the regime and not of the method class.

Two rows carry caveats rather than results. S3GM diverges here (1.019, worse than predicting the training mean) under its published predictor–corrector guidance at this benchmark’s parameter budget; we report it as a documented sampler failure with its diagnostic figures in the artifact rather than dropping the row, since a benchmark that hides the arms

TABLE V. Cylinder cross-Reynolds breakdown of Table IV, over the same 50 canonical frames (25 per Reynolds number). $Re = 80$ is an interpolation in Re ; $Re = 250$ is an extrapolation past the training maximum of 200. Aggregate relative L_2 , their ratio, and the never-observed pressure channel. Interpolation rows return the training mean on p by construction, and their aggregates coincide to four decimals by coincidence, not by error — both are dominated by that $p=1.0$ column.

Method	aggregate			p (never observed)	
	$Re\ 80$	$Re\ 250$	ratio	$Re\ 80$	$Re\ 250$
DMF-Gen	0.192	0.250	1.30	0.102	0.154
SiT	0.284	0.373	1.31	0.258	0.291
Latent FM	0.168	0.215	1.28	0.143	0.188
S3GM (PC-DPS)	1.013	1.025	1.01	1.007	1.002
Senseiver	0.129	0.162	1.26	0.071	0.118
MLP-RBF	0.325	0.452	1.39	0.177	0.265
Geo-FNO	0.074	0.111	1.50	0.069	0.107
Gappy POD ($r=20$)	0.044	0.068	1.54	0.037	0.064
IDW ($k=8$)	0.667	0.742	1.11	1.000	1.000
Nearest neighbor	0.667	0.742	1.11	1.000	1.000
Training mean	1.000	1.000	1.00	1.000	1.000

that break is not measuring robustness. Latent FM is the one over-dispersed model in the suite (spread-error 1.26, 90% coverage 0.89 against a 0.90 target), which is what a calibrated-but-blurry posterior looks like and is consistent with its strong CRPS at moderate relative L_2 . DMF-Gen sits on the other side of unity (spread-error 0.57, coverage₉₀ 0.51), so the two generative routes miss calibration in opposite directions on the same flow — an argument for the post-hoc recalibration of Sec. VD rather than for either architecture.

Which held-out Reynolds number? The training set is $Re \in \{60, 100, 150, 200\}$, so $Re = 80$ is an interpolation in Reynolds number and $Re = 250$ an extrapolation past the training maximum, and averaging the two hides the more informative question. Splitting the same 50 frames along that line (Table V) gives three results. The ranking does not change: gappy POD leads at both Reynolds numbers, Geo-FNO is second at both, Senseiver third. Extrapolating costs every working method between $1.26\times$ and $1.54\times$, and the cost is largest for the most accurate rows — gappy POD $1.54\times$ (0.044 to 0.068), Geo-FNO $1.50\times$ — and smallest for rows already near the training-mean floor (interpolation $1.11\times$): the methods that exploit the training family’s structure hardest lose most when asked to leave it, which is the counterweight to the headline that classical methods win this regime. And the never-observed pressure is the most Reynolds-sensitive part of the reconstruction: six of the seven rows that recover p at all degrade faster on p than on their aggregate (gappy POD $1.73\times$ against $1.54\times$, Senseiver $1.66\times$ against $1.26\times$, DMF-Gen $1.51\times$ against $1.30\times$; SiT is the exception), as expected for a channel reconstructed entirely from cross-channel structure learned at the training Reynolds numbers. This also explains the scalar-recalibration failure of Sec. VD: a single multiplier fitted across an interpolation and an extrapolation is fitting two error scales at once.

B. The same flow, sensed from the surface

Every result above places sensors throughout the volume. Experiments usually cannot: a wind-tunnel model carries pressure taps on its skin, and the flow away from the body is exactly what is unmeasured. We therefore repeat the cylinder task with sensors confined to the 360 wall-adjacent cells, all three fields read at the taps and nothing observed off the body, keeping the flow, the held-out Reynolds numbers, the frames, the seeded-draw machinery and the training budget identical. Predictions were registered before the runs

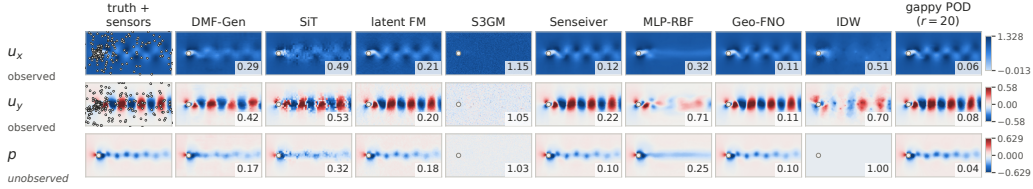


FIG. 2. Cylinder wake, held-out $Re = 250$ frame (val 300). Rows are the three reconstructed channels: u_x and u_y observed at 238 sensors per field (1%; circles on the truth panels) and p never observed. Columns are the truth and then one baseline each. Mesh-native methods (DMF-Gen, Senseiver, MLP-RBF, IDW, gappy POD) share one sensor draw; grid-native methods (SiT, latent FM, S3GM, Geo-FNO) receive the same count on the grid export, exactly as in Table IV. Generative columns show one posterior sample, not the ensemble mean; the number in each panel is that field’s single-sample relative L_2 in standardized units, so it sits above the table’s ensemble-mean error. The p row is the point: gappy POD infers the unobserved channel through the basis’s cross-channel correlations, S3GM’s sampler diverges on this dataset, and interpolation returns the training mean by construction.

and are reported whichever way they fell; two of the three failed.

The classical floor does not survive the change of geometry. Gappy POD with 20 modes wins the volume task outright at 0.056 (Table IV) and reaches only 0.689 from the full tap ring, while Senseiver improves from 0.145 to 0.103 and now leads by a factor of 6.7 over the classical anchor (Table VI). We had pre-registered the opposite risk — that POD might win here too, since the wake is genuinely low-rank — and it did not. The reason is that a modal basis needs its coefficients to be *identifiable* from the measurements, and a boundary ring samples a one-dimensional manifold on which the leading modes are nearly degenerate; low-rank structure in the field does not imply well-conditioned inference from an arbitrary sensor set. Interpolation fails completely, at 2.1 relative L_2 , worse than predicting the training mean, because it cannot extrapolate off the sensor manifold at all.

The rasterization cost is exact, not approximate. The 360-cell ring maps onto only 62 distinct cells of the 400×200 export, so every grid-locked method is blind to tap budgets above 62: Geo-FNO, SiT and latent FM return bit-identical errors at 64, 128 and 360 taps, to six decimal places, while the three mesh-native methods keep improving to 360. S3GM is grid-locked by the same argument; its 128- and 360-tap entries are marked $\dagger$ because they are inferred from the pool cap rather than re-measured. This is the capability matrix’s “requires resampling” column measured rather than asserted: the interface a method demands is a scientific property, not an implementation detail.

What the task does not show. Grid-locked is not the discriminator: Geo-FNO places second at 0.119, ahead of every point-native generative model. The surface geometry separates deterministic from generative methods, the reverse of what we expected from the argument that an under-determined far-wake posterior should reward a calibrated distribution. DMF-Gen is third at 0.245, behind both deterministic rows, and does not lead on CRPS either. We record the failed prediction.

C. Two-dimensional Kolmogorov turbulence

Sensors observe vorticity at 1% of the 256^2 points (655 sensors); learned models are scored on 50 evenly spaced frames of the eight held-out trajectories ($K=8$, identical canonical seeded draws across methods; classical floors on all 640). Table VII reports the full fleet alongside the floors, and Fig. 7 (left) the sensor-count sweep (Sec. VB). Three regime findings stand out.

First, *the 3D ranking does not transfer.* Geo-FNO wins relative L_2 outright at 0.417 — 14.5% better than the best generative model — while latent FM, the 3D aggregate winner,

TABLE VI. Cylinder *surface-to-field*: reconstruct the full wake from taps on the body alone, held-out $\text{Re} \in \{80, 250\}$, 50 Re -stratified frames. Columns are taps per field on the 360-cell wall ring; all three fields are read at the taps and nothing is observed off the body. $\dagger$ = identical to the 64-tap entry by construction: the ring rasterizes onto 62 grid cells, so grid-locked methods cannot represent a larger budget. Compare with Table IV, the same flow sensed in the volume, where the ranking is inverted.

Method	Rel. L_2 at n taps/field				CRPS
	32	64	128	360	
Senseiver	0.109	0.105	0.104	0.103	0.058
Geo-FNO	0.161	0.119	0.119	0.119	0.073
DMF-Gen	0.271	0.252	0.247	0.245	0.111
MLP-RBF	0.337	0.307	0.269	0.246	0.138
SiT	0.494	0.476	0.476	0.476	0.180
Latent FM	0.631	0.569	0.569	0.569	0.220
S3GM	0.897	0.891	0.891 $\dagger$	0.891 $\dagger$	0.468
Gappy POD ($r=20$)	1.223	0.919	0.781	0.689	0.306
IDW ($k=8$)	2.077	2.117	2.132	2.141	1.937
Nearest neighbor	2.159	2.153	2.148	2.144	1.940

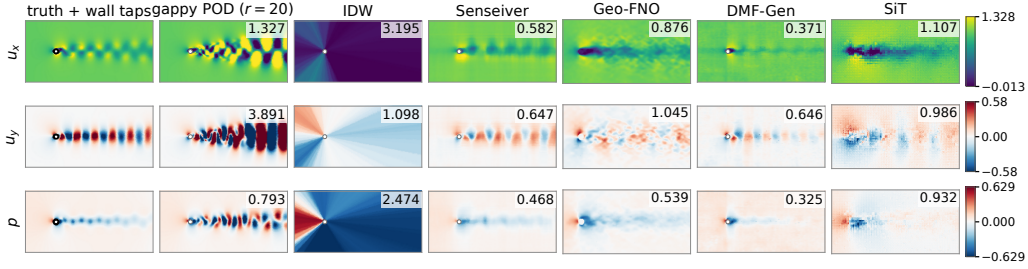


FIG. 3. Surface-only sensing on the held-out $\text{Re} = 250$ frame (128 taps per field, black dots on the body); everything off the body is inferred. Per-panel numbers are single-sample relative L_2 per field. Read against Fig. 2, the identical frame sensed in the volume: gappy POD, which wins there, is beaten several-fold here, and interpolation is worse than the training mean.

trails even the interpolation floor here (0.599 against 0.531) at the matched $\sim 50\text{k}$ -step budget, and DMF-Gen beats it on both relative L_2 and CRPS on all 50 paired snapshots. The remaining generative models land between them (DMF-Gen 0.487, SiT 0.494, S3GM 0.498); of these only DMF-Gen is ambient point-based here, since SiT and S3GM are run patch- and voxel-tokenized on this grid. The two deterministic point models split: MLP-RBF at 0.639 and Senseiver at 0.913, the latter barely inside the training-mean floor.

Second, *the accuracy and the probabilistic ranking disagree across method families, not within them*. The same Geo-FNO row that wins relative L_2 is beaten on CRPS by three of the four generative models (0.287 against S3GM 0.231, SiT 0.234, DMF-Gen 0.235; only latent FM, at 0.310, is worse), and the split is resolved on paired snapshots: Geo-FNO trails DMF-Gen on CRPS by 0.052 (95% CI [0.048, 0.057], all 50), while DMF-Gen and SiT are indistinguishable on it (0.001, $[-0.002, 0.004]$). This is the distortion-perception tradeoff of Sec. V A appearing as a family-level split rather than a curve: a conditional-mean estimator minimizes squared error and is then penalized by a proper score that rewards a calibrated distribution. Reporting either metric alone would name a different winner, which is why the benchmark requires both.

Third, the floors carry regime information of their own: gappy POD’s rank-80 training basis captures only 70.5% of the training-set energy — two-dimensional turbulence is not

TABLE VII. Kolmogorov 256^2 , trajectory-holdout evaluation (vorticity observed at 1% = 655 sensors, identical canonical seeded draws; learned rows: 50 evenly spaced held-out frames, $K=8$; classical floors: all 640). Deterministic rows carry $K=1$ and null dispersion, so their coverage cell is empty, their CRPS equals the MAE exactly, and their mean and single-sample columns coincide. Inference cost per field on one H100 (Geo-FNO’s budget-matched run: one GH200). Rows are generated from the canonical JSONs by `src/figs_pof/make_2d_tables.py`.

Method	Rel. L_2		CRPS	Cov. ₉₀	s/field	GB
	mean	sample				
DMF-Gen (NFE 4)	0.487	0.578	0.235	0.59	0.28	0.76
SiT (50 steps)	0.494	0.590	0.234	0.64	0.18	0.24
Latent FM (NFE 4)	0.599	0.731	0.310	0.64	0.03	0.19
S3GM (PC-DPS)	0.498	0.700	0.231	0.80	22.05	17.42
Senseiver	0.913	0.913	0.697	—	0.01	0.71
MLP-RBF	0.639	0.639	0.469	—	0.01	0.79
Geo-FNO	0.417	0.417	0.287	—	0.01	0.34
IDW ($k=8$) (per.)	0.531	0.531	0.375	—	—	—
IDW ($k=8$) (non-per.)	0.546	0.546	0.385	—	—	—
Nearest neighbor (per.)	0.582	0.582	0.371	—	—	—
Nearest neighbor (non-per.)	0.588	0.588	0.375	—	—	—
Gappy POD ($r=80$)	0.680	0.680	0.510	—	—	—
Training mean	1.000	1.000	0.753	—	—	—

low-rank in the POD sense, unlike the cylinder — and plain inverse-distance weighting beats it by 20% relative L_2 .

The winning row initially received four times the fleet’s training budget: the Kolmogorov budget was set in epochs at batch size 128, and Geo-FNO inherited the epoch count at batch size 32, hence 200k optimizer steps against the ~ 50 k of every other row (Sec. IIF). Retrained at exactly 50k steps with the configuration otherwise untouched, *Geo-FNO still wins*: 0.417 against DMF-Gen’s 0.487, SiT’s 0.494 and S3GM’s 0.498. The margin narrows from 21% to 14.5%; the four-fold budget was worth 0.032 relative L_2 to this architecture, a real effect and smaller than the gap it was suspected of manufacturing. The matched run is the row and the over-trained one an ablation, because a leaderboard should answer what a method achieves at the benchmark’s budget.

The paired analysis of Sec. IIE resolves Geo-FNO’s lead against the whole fleet (-0.071 versus DMF-Gen, CI $[-0.079, -0.063]$, $p < 10^{-14}$), whereas the three generative models behind it are not separable from each other: DMF-Gen versus SiT gives -0.006 with $p = 0.16$, SiT versus S3GM -0.004 with $p = 0.32$, and only the DMF-Gen–S3GM pair separates (-0.010 , $p = 0.008$), the expected intransitivity when three methods sit inside a 0.011 band. We report them as a tie group. Retraining every 2D method under two further seeds (7 and 1337) gives the same answer from the other side: the pooled training-seed standard deviation is 0.0041 relative L_2 on Kolmogorov and 0.0081 on the cylinder (S3GM is the most seed-sensitive row in both regimes, 0.0082 and 0.0163, consistent with its sampler fragility; SiT the least on Kolmogorov, 0.0006). Geo-FNO’s lead is seventeen seed standard deviations (its replicates were trained at the original $4\times$ budget, so its spread is measured there), while DMF-Gen–SiT and SiT–S3GM are one to one-and-a-half apart and cannot be called ordered; on the cylinder every adjacent gap exceeds the pooled spread by at least $3.7\times$, the tightest being latent FM against DMF-Gen at 0.030. Frame resampling and training-run resampling therefore agree on which comparisons this benchmark resolves.

The measurement operators of Sec. IID put the same ordering under corrupted observations (Table VIII). Gaussian noise at $\sigma = 0.1$ is invisible at this precision: no row moves by more than 0.007. At $\sigma = 0.3$ the four leading rows lose 6–12% (Geo-FNO $0.417 \rightarrow 0.457$, DMF-Gen $0.487 \rightarrow 0.525$, SiT $0.494 \rightarrow 0.555$, S3GM $0.498 \rightarrow 0.526$). A contiguous slab

TABLE VIII. Measurement operators on Kolmogorov at 1% sensing (655 sensors): ensemble-mean relative L_2 over the 50 canonical test frames ($K=8$ for generative rows, one forward pass for deterministic ones). Every method receives bit-identical corrupted observations (Sec. IID); σ is Gaussian noise in standardized units, and the slab removes one contiguous region holding 25% of the sensors. Last column: occluded over clean. Geo-FNO is the budget-matched run.

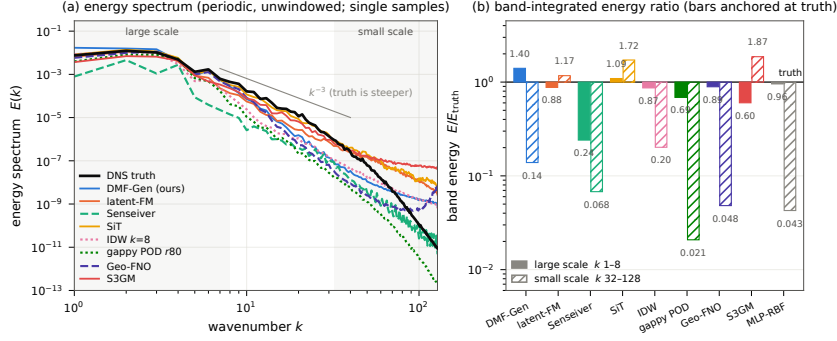
Method	clean	$\sigma=0.1$	$\sigma=0.3$	25% slab	slab/clean
Geo-FNO	0.417	0.421	0.457	0.554	1.33
DMF-Gen	0.487	0.492	0.525	0.660	1.35
SiT	0.494	0.501	0.555	0.709	1.44
S3GM (PC-DPS)	0.498	0.503	0.526	0.651	1.31
Latent FM	0.599	0.601	0.608	0.695	1.16
MLP-RBF	0.639	0.640	0.645	0.763	1.19
Senseiver	0.913	0.913	0.915	0.920	1.01

removing a quarter of the sensors costs those rows far more, 31–44%, because it leaves a region with no nearby data rather than thinning the data everywhere. Geo-FNO keeps its lead under every operator (0.554 occluded, against 0.651 for the next row). Below it the clean tie group reshuffles: SiT degrades most ($1.44\times$) and ends level with latent FM (0.709 against 0.695), which it led by 0.105 on clean data, while S3GM (0.651) and DMF-Gen (0.660) stay 0.009 apart — about two clean-fleet seed standard deviations — and we do not rank them. The rows that barely notice any operator are the ones with the highest clean error: Senseiver moves from 0.913 to 0.920 under occlusion, and latent FM and MLP-RBF lose 16–19%. Read alone, the last column of Table VIII would rank the least accurate model as the most robust; insensitivity to losing sensors is only a virtue in a model that was using them. Channel dropout is not in the table: Kolmogorov observes a single channel, so removing it would leave nothing to condition on.

Spectrally the fleet separates differently again (Fig. 4). Measured against the truth spectrum of the same frame over a small-scale band ($k = 32\text{--}128$), the samplers that inject texture overshoot — latent FM carries 1.17 of the true energy, SiT 1.72 and S3GM 1.87 — while DMF-Gen carries 0.139 and the deterministic rows almost none (Geo-FNO 0.048, Senseiver 0.068, MLP-RBF 0.043, gappy POD 0.021). The ordering is the reverse of the 3D result, where the latent decoder truncated the smallest scales and the ambient point model retained them (Sec. VE). Ratios above unity are errors too: small-scale energy in excess of the truth is spurious texture rather than resolved structure, and the nearest-neighbor interpolant reaches 2.55 in the same band through piecewise-constant blocking. The accuracy leader carries almost none of that band — Geo-FNO wins relative L_2 with 0.048 of the true small-scale energy — so the distortion-perception split of Sec. VA reappears here as a spectral one. The samplers ran at their own budgets (DMF-Gen and latent FM at NFE 4, SiT at 50 steps, S3GM at 200), and a coarse solver budget can bias a sample smooth, but that is not the explanation here: re-sampling DMF-Gen on the same frame, sensors and seed at SiT’s 50 steps raises its small-scale energy only from 0.139 to 0.185 of the truth (0.174 at 16 steps), against SiT’s 1.72 at the same budget, while its large-scale excess grows (1.40 to 1.67) and its single-sample error worsens (0.71 to 0.78). The 2D small-scale deficit is architectural, not a coarse-solver artifact.

D. Three-dimensional isotropic turbulence: the full fleet

The 3D turbulence benchmark is the controlled comparison: a periodic-topology regular grid is native terrain for voxel, patchified, and latent-grid methods, so every family competes at full strength. Sensors observe U_x and U_z at 1% of 1.95×10^6 points per channel; U_y and



Kolmogorov flow on $[0, 2\pi]^2$, held-out frame (val 256), 655 sensors (1%), 256^2 vorticity. The domain is PERIODIC, so the plain FFT is exact and NO WINDOW is applied here — unlike the Hann-windowed 3D spectra elsewhere in this paper, whose cutouts are non-periodic. Single posterior samples throughout (an ensemble mean would destroy the small scales by construction); sampler budgets are each method’s own and are NOT matched (DMF-Gen NFE=4, latent-FM NFE=4, SiT NFE=50, S3GM NFE=200), which biases the small-scale band of the lower-budget samplers smooth.

FIG. 4. Kolmogorov spectral fidelity at 655 sensors (1%), held-out frame, one reconstruction per method. (a) Shell-averaged energy spectra $E(k) = \sum_{\text{shell}} |\hat{\omega}(k)|^2 / 2k^2$ on integer shells to the Nyquist limit. The field occupies the doubly periodic box $[0, 2\pi]^2$ and the saved frame is a full period, so no window is applied here, unlike the Hann-windowed 3D spectra of Fig. 8, whose cutouts are non-periodic sub-blocks; the estimators differ because the boundary conditions differ. The k^{-3} line is a reference slope, not a fit: the DNS is steeper at this Reynolds number. (b) Band-integrated energy relative to truth over $k = 1-8$ and $k = 32-128$, on a log axis anchored at unity so deficit and spurious excess both read as error. All generative entries are single posterior samples; MLP-RBF, a ninth series past the validated eight-color palette, appears in (b) only. Sampler budgets are each method’s own; the NFE-matched control in the text shows the budget does not explain the ordering.

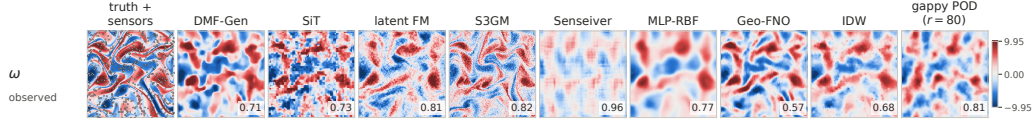


FIG. 5. Kolmogorov 256^2 , canonical held-out frame (val 256). Vorticity is the only channel and is observed at 655 sensors (1%; dots on the truth panel), with the identical draw in every column. Columns and per-panel numbers as in Fig. 2. Gappy POD is visibly over-smoothed — two-dimensional turbulence is not low-rank — the generative samples carry small-scale texture the deterministic rows lack, and the patch tokenizer’s block structure is directly visible in the SiT panel.

p are never observed. Table IX reports per-channel accuracy and probabilistic scores for the full fleet on all 50 held-out snapshots of the spatially disjoint cutout under canonical seeded draws; Table X reports the matched cost accounting.

No method leads every column. SiT-point posts the best observed-channel errors (0.143/0.132) at roughly 150s per field — 239 sequential 8,192-token chunks at 32 steps — and its single samples carry chunk-incoherent speckle that the ensemble mean averages away. Latent flow matching leads the aggregate, the unobserved channels, and CRPS: its convolutional decoder emits globally structured fields even where no sensor constrains them, at the dissipation-range price quantified in Sec. V E. Non-periodic inverse-distance weighting — a training-free interpolant — scores 0.174 on U_x , ahead of every learned row except SiT-point (DMF-Gen 0.176, FNO3D 0.188, latent FM 0.248), so at 1% density the learned and generative payoffs concentrate where the problem is ill-posed — unobserved channels, sparser sensing (Fig. 7) — and in sampling cost (DMF-Gen reaches its accuracy in 4 network evaluations against 32–1000 for the guided-diffusion and transformer baselines). DeepONet’s near-constant predictions are the structural consequence of global mean pooling in its branch; the labeled DeepONet++ arm (structured pooling) improves U_x from

TABLE IX. 3D isotropic turbulence, spatially disjoint evaluation region (1.95M points; U_x, U_z observed at 1% per channel, U_y^*, p^* unobserved; canonical seeded sensor draws, all $n=50$ held-out snapshots, $K=8$ ensembles at each method’s listed sampler setting). Per-channel relative L_2 of the ensemble mean is primary; $\text{agg}^\dagger$ averages all four channels and is flagged, not headlined. Unobserved-channel error ≥ 1.0 is worse than predicting the training mean; values near 1.0 across the fleet reflect a shared identifiability limit of point-conditioned methods (Sec. V C). Spread–error and coverage ideals are 1.0 and 0.90; for deterministic methods CRPS reduces to MAE and dispersion is undefined. Step budgets are as-run, not equalized (Table X). Labelled improvement arms (Senseiver capacity, DeepONet++, CoNFILd repairs, S3GM normalized guidance) are reported in the artifact. ‡ = guidance deviates from the published setting, which is why the upstream-faithful S3GM row is reported separately above it rather than replaced by the working one; the normalized-guidance coefficients were fitted on the tuning split only, and its tune/test difference is below 0.001 relative L_2 .

Method	U_x	U_y^*	U_z	p^*	$\text{agg}^\dagger$	CRPS	Spr/Err	Cov. ₉₀
DMF-Gen (NFE 4)	0.176	1.048	0.182	0.964	0.593	0.291	0.63	0.54
Latent flow matching	0.248	0.727	0.257	0.646	0.469	0.244	0.68	0.56
SiT-point ($N=32$)	0.143	0.968	0.132	0.937	0.545	0.338	0.75	0.58
FNO3D (same framework, NFE 4)	0.188	0.904	0.191	1.061	0.586	0.269	0.94	0.63
CoNFILd (published prior, DPS 1000)	0.409	0.751	0.374	1.007	0.635	0.371	0.82	0.57
CoNFILd (capacity-matched)	0.488	1.089	0.420	1.487	0.871	0.562	0.62	0.45
CoNFILd (384-d)	0.524	0.604	0.451	1.401	0.745	0.480	0.69	0.50
Gen4Turb (anneal, 32-step)	0.224	0.996	0.455	1.005	0.670	0.407	0.37	0.33
S3GM (upstream guidance, $N=200$)	diverges at 3D scale (rel- L_2 8.9×10^5); see text							
S3GM (retuned guidance) ‡	0.362	0.992	0.350	1.024	0.682	0.368	0.48	0.42
S3GM (normalized guidance) ‡	0.260	0.980	0.254	1.018	0.628	0.343	0.52	0.46
Senseiver (deterministic)	0.361	1.077	0.380	0.978	0.699	0.479	—	—
DeepONet (deterministic)	0.593	0.972	0.652	0.985	0.800	0.570	—	—
Nearest neighbor (KD-tree)	0.209	1.000	0.219	1.000	0.607	0.406	—	—
IDW ($k=8$, non-periodic)	0.174	1.000	0.181	1.000	0.589	0.396	—	—
Gappy POD ($r=80$)	0.523	1.306	0.540	1.443	0.953	0.673	—	—
Training mean	1.000	1.000	1.000	1.000	1.000	0.786	—	—

TABLE X. Cost accounting for the 3D-turbulence fleet (H100-SXM 80 GB). Steps are as-run, not equalized: every external baseline received at least DMF-Gen’s 48k optimizer steps except SiT-point (wall-clock-matched, 38.5k); CoNFILd stages train on a wall-clock budget. “(d)” = derived from evaluation-job wall time rather than a dedicated benchmark, and therefore an *upper bound*: it includes data loading, host–device transfer and metric computation, which the dedicated numbers exclude, and per-call setup that a dedicated benchmark amortizes. The marked rows are the slow samplers, where the sampling term dominates the overhead, so the comparison the column supports (orders of magnitude between families) is unaffected.

Method	Params	Opt. steps	Train s/step	Train GPU-h	Train peak GB	Infer s/field
DMF-Gen (NFE 4)	6.51M	48k	0.574	7.7	53.6	~17 (d)
Latent flow matching	6.68M	209k+95k	0.238	13.9	3.6	0.019
SiT-point ($N=32$)	6.57M	38.5k	1.81	19.4	26.8	~150 (d)
FNO3D (NFE 4)	6.73M	79.9k	0.766	17.0	48.6	0.097
CoNFILd (pub. prior)	118.9M ^w	wall	35.5/ep	~19	4.7	~130 (d)
Gen4Turb (anneal)	6.16M	105k	0.598	17.4	47.4	1.65
S3GM (JHU-tuned)	6.35M	105k	0.635	18.5	26.3	~46
Senseiver	6.42M	79.2k	0.776	17.0	22.4	<1
DeepONet	6.51M	82.0k	0.29	6.7	15.7	0.23
IDW $k=8$ (CPU)	—	—	—	—	10 (RSS)	0.68
Gappy POD $r=80$ (CPU)	—	45 s fit	—	—	10 (RSS)	0.14

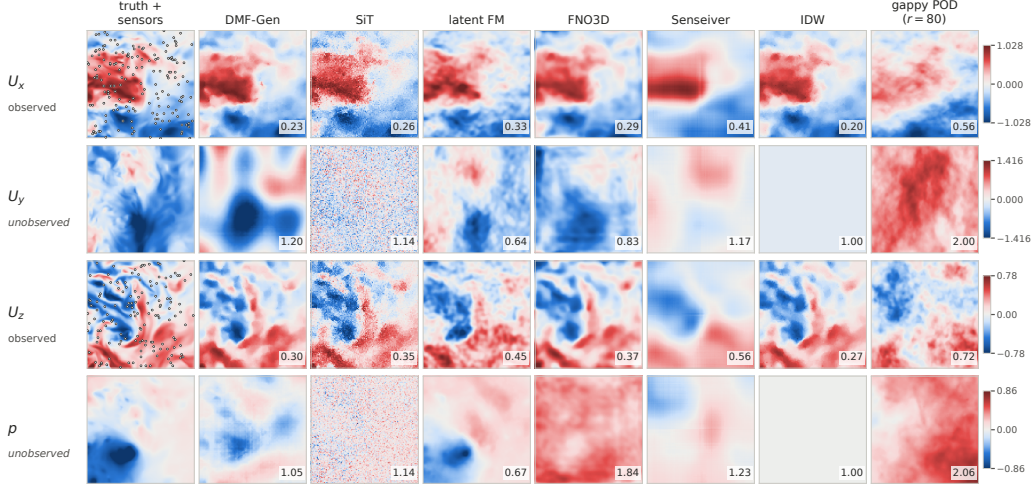


FIG. 6. 3D isotropic turbulence, held-out cube (frame 153), z -midplane. Rows are all four channels: U_x and U_z observed at 19,531 sensors each (1%; in-plane sensors shown as circles), U_y and p never observed; the sensor set is identical in every column. Per-panel numbers are single-sample relative L_2 over the whole cube. The unobserved rows are the identifiability wall of Sec. VC: latent FM, whose decoder imposes global field structure, is the only method below 0.7 on either unobserved channel and places the low-pressure core, while interpolation returns the training mean, gappy POD extrapolates away from the truth, and the point-token sampler emits incoherent noise. CoNFILd, S3GM (which diverges at this scale), DeepONet++ and Gen4Turb are omitted from the gallery.

0.593 to 0.512 and remains behind the point-attention family.

S3GM’s three rows must be read together. At the published guidance coefficients ($\alpha_{\text{case}}=0.5$, $\beta=0.4$) the sampler diverges at this scale by five orders of magnitude, each coefficient independently — a property of the guidance at 1.95M points, not a porting defect. Reducing both coefficients by two orders of magnitude stabilizes the sampler at 0.682 aggregate; a normalized guidance whose coefficients are fitted on the tuning split reaches 0.628 with observed-channel errors of 0.260 and 0.254. Both are labeled deviations from the published method, neither can be quoted as “S3GM” without qualification, and neither closes the unobserved-channel gap (0.98–1.02) that no method in the fleet closes.

The top of this table is partly a tie. Applying the paired analysis of Sec. IIE to the canonical $n=50$ JSONs, latent FM’s lead is resolved against both of the rows immediately behind it (against FNO3D -0.118 , CI $[-0.147, -0.088]$; against DMF-Gen -0.124 , CI $[-0.147, -0.102]$; both $p < 10^{-9}$), while FNO3D and DMF-Gen are not separable from each other (-0.007 , CI $[-0.027, +0.015]$, $p = 0.33$) despite being reported as different rows. A ranking that reads the third and fourth entries as ordered would be over-reading the benchmark’s resolution at this n .

Two limits bound what this table can support. First, only the three rows above could be compared frame by frame: the SiT-point, Senseiver, Gen4Turb and DeepONet rows have no canonical per-snapshot payloads in the artifact set, so their positions are point estimates without a paired interval and we do not claim any of them is separated from a neighbor. (A fourth CoNFILd arm, with strict 2048-point conditioning, is omitted altogether: its files aggregate to 0.941 relative L_2 but carry no protocol stamp, so the artifact cannot confirm that it saw the canonical sensor draw.) Second, the seed error bar covers the two 2D regimes only; the 3D rows are single training runs, so a 3D comparison called resolved is resolved by evaluation-frame resampling alone.

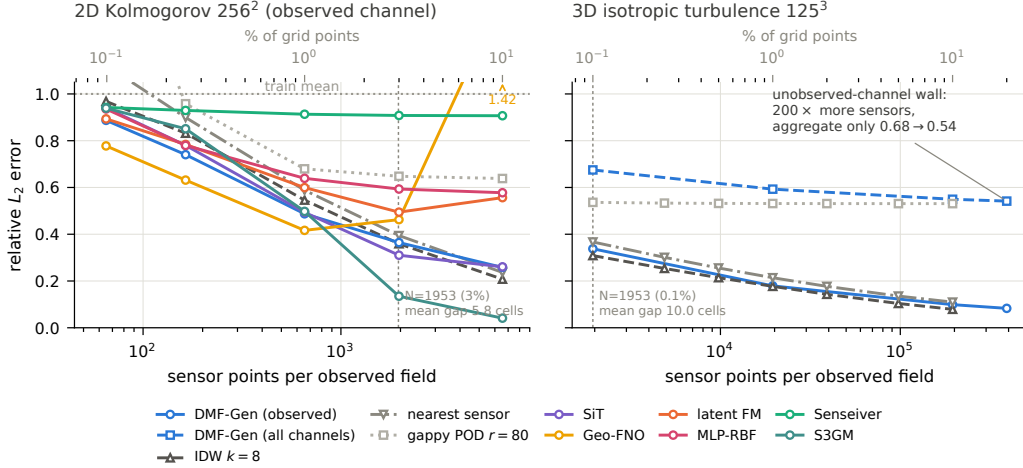


FIG. 7. Reconstruction error versus the number of sensor points, in 2D (left, Kolmogorov 256^2 , full learned fleet) and 3D (right, JHU 125^3 cross-cube; solid = observed-channel rel- L_2 , dashed = aggregate over all four channels). Colored = learned; gray = classical floors. The same sensor count that leaves a 5.8-cell mean gap in 2D leaves a 10-cell gap in 3D. In 3D every observed-channel error falls steadily with sensor count while the aggregate saturates near 0.54, because the unobserved channels do not identify (Sec. VC); gappy POD is flat at every density. In 2D the fleet fans out (Sec. VB): S3GM passes every other method from 1965 sensors on, Geo-FNO degrades past the canonical density (1.42 at 10%, marked off-scale), and Senseiver is nearly flat. Learned rows: canonical protocol, 50 held-out frames, K as in Table VII, identical sensor draws across methods at every density; classical curves on the same draws.

V. CROSS-CUTTING ANALYSES

A. The distortion–perception tradeoff, operationally

Relative L_2 of the ensemble mean is minimized by the posterior mean — by construction it rewards collapse toward a smooth conditional average — while proper scores and physics metrics reward calibrated diversity and sharp members. The benchmark measures the tension directly through the sampler’s operating point: for the rectified-flow methods, two integration steps give the best mean-field error and the sharpest single samples, four steps are the CRPS optimum, and dispersion rises monotonically with further steps at flat CRPS (spread–error 0.65 at NFE 2 to 0.92 at NFE 16 for the $\lambda_{sp}=0$ DMF-Gen variant). A spectral training penalty moves along the same frontier: it buys single-sample sharpness and inertial-band fidelity at a measured cost in dispersion. The practical rule the benchmark enforces: never rank generative reconstruction methods on mean-field error alone, and attach the sampler operating point to every probabilistic number.

B. Sensor density: where learning pays

The 2D panel of Fig. 7 settles a question the single-density tables cannot: *over what range of sensor budgets does learning pay for itself?* Sweeping the Kolmogorov fleet over $\{65, 164, 655, 1965, 6554\}$ sensors — 0.1% to 10% of the grid, canonical draws at every density — bounds that range on both sides for most of the fleet, and shows that the bound is itself method-dependent. At the sparse end learning wins clearly: at 65 sensors Geo-FNO reaches 0.778 and DMF-Gen 0.887 while IDW manages only 0.968 and gappy POD (1.82) and nearest-sensor (1.12) are both worse than predicting the training mean. At the dense end the fleet splits. Plain inverse-distance interpolation reaches 0.208 at 6554 sensors and

beats six of the seven learned rows, the best of which are DMF-Gen at 0.258 and SiT at 0.261; for those six the crossover falls between 1965 and 6554 sensors, so their case for learning lives below a few percent sensing — the regime this benchmark targets, and the one uniform-downsampling protocols leave (Sec. V G).

The seventh row inverts that picture. S3GM, the guided video-diffusion prior, is near the bottom of the fleet at 65 sensors (0.939) and at the top from 1965 sensors on: 0.135 there against SiT’s 0.310, and 0.041 at 6554, five times better than interpolation. Its curve is the steepest in the benchmark ($23\times$ over the sweep against $3.4\times$ for DMF-Gen), consistent with posterior-sampling guidance whose likelihood term strengthens with the measurement count until it overrides the prior — a mechanism we report as consistent with the data, not established by it. The gain has two prices: 22s per field against 0.28 for DMF-Gen (Table VII), and a stated spread that does not shrink with the error (spread-error rises from 1.07 at 1% to 2.56 at 10%, coverage₉₀ from 0.80 to 0.98), so the most accurate reconstruction in the sweep is also its most over-dispersed.

The fleet does not share a slope, and the exceptions are diagnostic. Five rows fall monotonically over the $100\times$ range (S3GM, DMF-Gen, SiT, MLP-RBF, Senseiver). Two do not: Geo-FNO is best at the canonical density and degrades from there — 0.417 at 655 sensors, 0.463 at 1965 and *1.42* at 6554, worse than the training mean — and latent FM turns the same corner later and more gently ($0.495 \rightarrow 0.557$ past 1965 sensors). Both are the rows whose conditioning path resamples the sensors onto a fixed representation, a grid or a latent code, so sensors added past the point where that representation saturates add conditioning mismatch rather than information. A single-density leaderboard cannot see this: at 1% Geo-FNO is the best learned method on the regime, and nothing in that number warns that ten times the instrumentation would make it three times as bad. Senseiver’s near-flatness is the same lesson from the other direction: it extracts at 65 sensors essentially all it will ever extract. *Which* method pays for learning is itself density-dependent, which is why the sweep is reported as a first-class result.

C. Unobserved channels: a fleet-wide identifiability limit

On 3D turbulence, no method reconstructs the unobserved U_y below 0.60 relative L_2 , and most sit near or above 1.0, worse than the training mean (Table IX). The two partial exceptions (latent FM at 0.727, CoNFILD-384d at 0.604) are the methods whose decoders impose the strongest global field structure, and both pay for it elsewhere. This is not a per-method defect: at 1% point sensing on two velocity components, the instantaneous realization of a third component of 3D turbulence is close to unidentifiable, and only its statistics are recoverable. The cylinder’s never-observed pressure makes the contrast quantitative: where 3D turbulence compresses every method into the 0.90–1.00 band, the same measurement on the cylinder spreads them over an order of magnitude — gappy POD 0.043, Geo-FNO 0.088, Senseiver 0.095, DMF-Gen 0.128, SiT 0.274, interpolation 1.000 by construction (Table IV). Identifiability is a property of the flow and the sensing geometry, not of a method class, and a benchmark that reported only 3D would license the wrong conclusion.

The observe- u -only arm makes the point sharper still. Retraining on u_x sensors alone — so that u_y joins p as an unobserved channel, and the total sensor budget halves from 476 to 238 — leaves Senseiver’s u_y error unchanged at 0.231, against 0.240 when u_y was measured directly at 1%. Removing an entire observed channel costs it nothing, because at this Reynolds number the shedding phase inferred from u_x determines the rest of the field; its aggregate moves only from 0.145 to 0.148. DMF-Gen is not indifferent in the same way: its u_y error rises from 0.353 to 0.438 and its aggregate from 0.221 to 0.250. The contrast with Table IX, where no method recovers an unobserved channel of 3D turbulence at any budget, is the cleanest statement the benchmark makes about identifiability: whether a channel is recoverable is set by the flow, and *how efficiently a method exploits that recoverability* is what separates architectures. Reporting only the aggregate would hide both halves.

D. Calibration: characterized, then repaired

Almost every scored ensemble in the fleet is under-dispersed at the headline operating point (spread–error 0.37–0.94 against the ideal 1.0; coverage 0.33–0.63 against 0.90), consistent with every scored precedent in adjacent literatures.^{19,20} The two exceptions are instructive rather than contradictory: cylinder latent FM is over-dispersed (spread–error 1.26, coverage 0.89) and Kolmogorov S3GM sits just past unity (1.07, coverage 0.80). Both are latent- or guidance-driven samplers whose spread is set by a mechanism other than the observation density, and neither is more accurate for it — latent FM’s relative L_2 on the cylinder (0.192) is twice the Geo-FNO row’s. Calibration and accuracy are separately earned, which is the case for reporting both. For the best-characterized model the miscalibration has a low-dimensional mechanism: predictive spread follows a distance-to-nearest-sensor profile with a hard floor (≈ 0.09 in standardized units), the profile is density-invariant, and the sampler step count rescales it wholesale ($\approx 1.5\times$ from NFE 4 to 16). That structure makes post-hoc repair tractable, and the benchmark ships two estimators fitted strictly on the tuning split and frozen on test: a per-density scalar spread multiplier, and conformalized quantile scaling binned by distance-to-nearest-sensor, which carries a finite-sample marginal coverage guarantee on exchangeable points. The identical wrapper is applied to the generative baselines, so the repaired comparison is two-sided. The repair is post hoc, rescales stated uncertainty without sharpening the spread floor, and every repaired number carries its operating point.

The scalar estimator, fitted on the tuning half and evaluated frozen on the disjoint test half, moves the test spread–error ratio to within a few percent of unity wherever the miscalibration is stationary. On 3D turbulence the DMF-Gen row is repaired at every density — $0.436 \rightarrow 1.001$ at 0.1% sensing ($\alpha=2.30$), $0.550 \rightarrow 1.017$ at 1% ($\alpha=1.85$), and $0.949 \rightarrow 1.016$ at 10% ($\alpha=1.07$) — and the multiplier falls toward one as sensing densifies, exactly as the distance-to-sensor mechanism predicts. The same wrapper applied to latent FM on the identical protocol gives $0.706 \rightarrow 0.993$, and on Kolmogorov it repairs all four generative rows ($0.674 \rightarrow 0.975$ for DMF-Gen, $0.691 \rightarrow 0.985$ SiT, $0.729 \rightarrow 0.981$ latent FM). It also corrects in the other direction: the over-dispersed Kolmogorov S3GM row is shrunk rather than inflated ($\alpha=0.95$, $1.046 \rightarrow 0.996$), which is the behavior a calibration repair should have and which an inflation-only heuristic would miss.

The cylinder is the counterexample. There the repair does not transfer: DMF-Gen overshoots ($0.582 \rightarrow 1.112$) and latent FM is essentially untouched ($\alpha=1.01$, $1.189 \rightarrow 1.200$), because the held-out set spans two Reynolds numbers whose error and spread scales differ, so a single scalar fitted on one half is the wrong functional form for the other. A scalar repair assumes the miscalibration is stationary across the reporting set; when that set is heterogeneous by construction the fix must be conditioned on the regime variable, which is why we report the fit rule, the split and the before/after pair for every row. (The cylinder’s canonical frames are all even-indexed, so its tune/test halves are taken by position in the frame list; both halves draw from both held-out Reynolds numbers.) The conformalized estimator repairs the quantity the scalar one cannot. Fitting a per-distance-bin conformal quantile on the tuning snapshots and freezing it on test, 90% coverage on U_x moves from 0.403 to 0.895 at 0.1% sensing, 0.560 to 0.900 at 1%, and 0.755 to 0.904 at 10%, with the unobserved channels behaving the same way (0.409 to 0.898 on p); the same holds at every nominal level we report. That is the finite-sample split-conformal guarantee doing exactly what it promises, on five million held-out points. The two estimators also agree about how much repair the model needs: the raw coverage deficit shrinks monotonically with sensing density (0.40, 0.56, 0.76) exactly as the scalar multiplier falls toward one (2.30, 1.85, 1.07), which is the signature of both instruments measuring a single underlying miscalibration rather than two unrelated defects.

One estimator caveat: the conformal fit bins by distance to the nearest sensor, and at 10% sensing that distribution concentrates enough that equal-count bin edges coincide and a bin can come up empty on the tuning half. An empty bin has no quantile; emitting zero there would produce zero-width intervals and destroy coverage for exactly the points

that land in it. We collapse duplicate edges and fall back to the pooled quantile for any empty bin, which is conservative rather than optimistic, and the reported bin count is the post-collapse one.

Reporting both estimators together is more informative than either alone, because they disagree in a diagnostic way. After the conformal repair the *coverage* is exact but the spread-error ratio overshoots to 1.2–1.6, while the scalar multiplier sets spread-error to 1.0 and leaves coverage short. Neither is wrong: a scalar matches the second moment, a conformal quantile matches the tail probability, and the two coincide only for Gaussian residuals. The gap between them is therefore a measurement of how non-Gaussian the reconstruction error is — heavy-tailed enough that the interval achieving 90% coverage is half again wider than the standard deviation would suggest. A benchmark that reported a single calibration number would hide that, and a practitioner who needs calibrated intervals should take the conformal one while a practitioner who needs an error bar on the mean should take the scalar.

The same script, applied unchanged to the latent flow-matching dumps (same frames, sensor draws, K and tuning/test split), makes the comparison two-sided. Latent FM starts less under-covered than DMF-Gen at sparse sensing and more at dense: raw 90% coverage on U_x is 0.553, 0.654 and 0.661 at 0.1%, 1% and 10%, against 0.403, 0.560 and 0.755, so its deficit barely responds to sensing density above 1%. The conformal repair brings it to nominal as well — 0.897–0.898 on the observed channels at every density — and slightly past nominal on the unobserved ones (0.911–0.912 on U_y , 0.916–0.924 on p), which is the conservative side. The estimator disagreement measured above separates the two models: after the repair, latent FM’s spread-error ratio on the observed channels is 1.11–1.12 against 1.19–1.57 for DMF-Gen, so its observed-channel residuals are closer to Gaussian, while on pressure it is the heavier-tailed of the two (1.33–1.37 against 1.19–1.20). Neither model’s intervals are usable as stated; both are after a split-conformal step fitted on held-out tuning frames.

E. Spectral fidelity splits by band

Windowed shell spectra of single samples (Fig. 8) separate the generative families where pointwise error cannot. On 3D turbulence the latent route holds more inertial-range energy than the ambient point-cloud route (0.62 vs. 0.41 of DNS energy on the observed U_x , $k \in [8, 31]$) but collapses in the dissipation band, retaining 0.07–0.09 of DNS energy over $k \in [32, 45]$ against the ambient model’s 0.40–0.46 — the autoencoder spectral floor^{31,32} appearing where it was predicted, a factor-of-five separation invisible to relative L_2 . Neither family reproduces the DNS spectrum from 1% sensors: the ambient model’s high- k content flattens into a broadband-noise plateau on the unobserved pressure channel (3.5–4.8× DNS energy), and on channels no sensor constrains its samples stay near its spectrally empty source prior (inertial ratio 0.03–0.06 on U_y , versus 0.67 for the latent decoder that paints texture everywhere). Member-level spectra, windowed estimation, and per-channel reporting are all load-bearing; ensemble-mean spectra and unwinded estimators each silently manufacture different conclusions.

F. Computational cost and scaling

Cost separates the fleet along two axes. At the benchmark resolution (Table X), total training cost to a finished model spans 7–19 GPU-hours across learned methods — comparable — but inference spans five orders of magnitude, from 19 ms (latent decode) to minutes (chunked token generation, DPS optimization), and peak training memory spans 3.6–54 GB. Against resolution (Fig. 9), the fleet splits into two families: grid computation (voxel diffusion, convolutional autoencoders) scales cubically with measured out-of-memory walls at 320^3 (training, batch 1) and 640^3 , while point computation (attention over sensors with

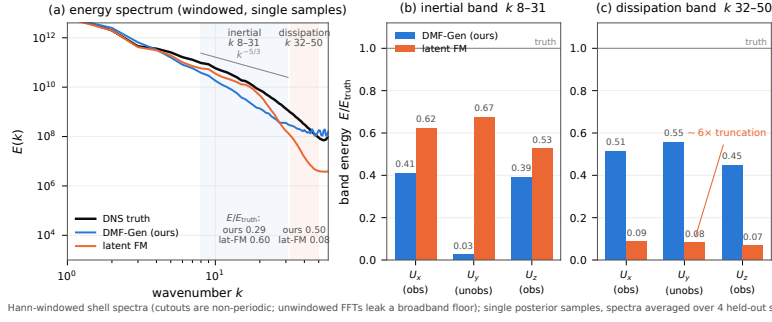


FIG. 8. Turbulence statistics of single posterior samples on the held-out 3D region (4 snapshots, NFE 4, Hann-windowed; ensemble means are deliberately not shown — averaging destroys small scales by construction). Left: shell-averaged energy spectra — the latent baseline tracks the DNS through the inertial range then falls away sharply in the dissipation band (autoencoder floor); the ambient model sits lower in the inertial range and flattens into a noise plateau at high k . Middle: longitudinal structure functions. Right: velocity-gradient PDFs.

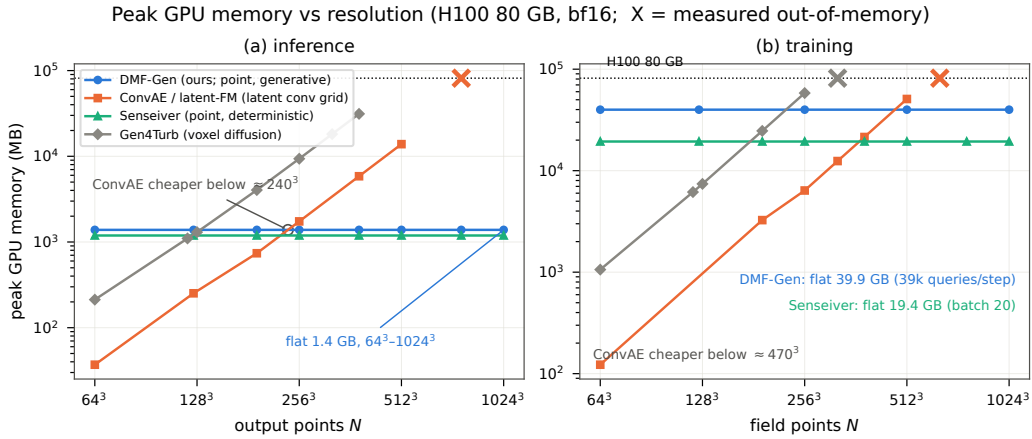


FIG. 9. Measured peak memory versus output resolution (H100, bf16, log-log; OOM markers are measured failures, not extrapolations) for full-field sampling (left) and training steps (right). Grid methods (voxel diffusion, convolutional AE) scale cubically and fail at 320^3 and 640^3 ; point methods (DMF-Gen, Senseiver) are flat to 1024^3 (10^9 points). Annotation: fraction of the field supervised per training step — 100% forced for grid methods, a fixed 39k-point budget for the subsampled point method.

chunked queries) is flat to 1024^3 in both training and inference memory. The converse is that grid computation is cheaper below the crossover ($\sim 240^3$ for inference: a convolutional decode of a 125^3 field costs 6 ms where chunked point evaluation costs seconds). No family dominates; the benchmark's contribution is that the walls and crossovers are measured, so a practitioner can place their discretization on the axis before choosing a method family.

Figure 10 makes the same point from the accuracy side: on the cost-accuracy plane alone the latent baseline dominates every method including DMF-Gen, and the case for ambient point-cloud generation rests entirely on the axes the plane omits — resolution scaling, discretization freedom and dissipation-range fidelity, measured here, and behavior under corrupted operators (Table VIII).

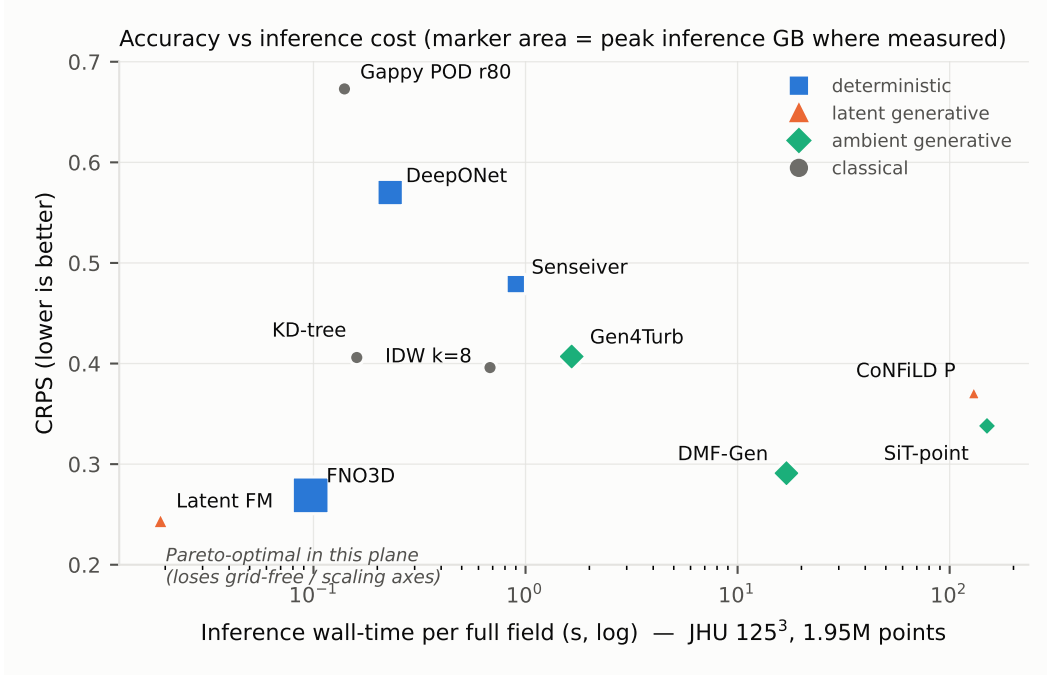


FIG. 10. Probabilistic accuracy (CRPS, JHU cross-cube, $n=50$) against inference wall-time per full 125^3 field, marker area proportional to peak inference memory where measured (Table X). At the benchmark resolution the latent flow-matching baseline Pareto-dominates this plane outright — fastest and lowest CRPS — which is precisely why a single cost–accuracy plot is an insufficient basis for method choice: the axes it cannot show (Fig. 1; the scaling walls of Fig. 9; the dissipation-band ratios of Sec. V E) are where the grid-locked methods lose. S3GM is absent from this plane: its upstream sampler diverges at 3D scale, and its two labeled retuned arms are reported in Table IX rather than plotted as if they were the published method (Sec. IV D).

G. Why benchmark numbers look worse than the literature’s — protocol, not architecture

Readers who compare our tables and galleries against published reconstruction papers — including papers evaluated on the very same Kolmogorov data⁸ — will notice that every method here, including the strong ones, scores and *looks* worse. We consider this a central result of the benchmark, because each contribution to the gap is a protocol choice, and we measure them separately.

(i) *Splits — a correctness requirement, not the explanation.* Turbulence frames are strongly autocorrelated (on the 3D data, $r = 1.00$ at unit separation and 0.67 at separation 100), so a random-in-time split places every evaluation frame next to a training frame, and we expected the leak to be large. We measured it with the same probe in both regimes: frozen checkpoints re-scored on frames they were trained on, which bounds from above what any random-in-time split could return. On 3D turbulence, 50 frames of the training cubes chosen by the canonical selection rule and sensed by the canonical seeded draws give latent FM 0.453 aggregate relative L_2 against 0.469 on the held-out cube (observed U_x 0.240 against 0.248 and U_z 0.277 against 0.257; unobserved U_y 0.665 against 0.727) and FNO3D 0.586 against 0.586, while the training-free interpolation control is unchanged (IDW 0.589 against 0.589), so the training frames are not easier. DMF-Gen is the one row with a material in-sample gain, 0.515 against 0.593 aggregate, and it sits almost entirely on the never-observed pressure (0.678 against 0.964) while its observed channels move by 5% or less (U_x 0.167 against 0.176, U_z 0.190 against 0.182): memorization buys something only where the sensors constrain nothing, and even there the channel stays unrecovered. On Kolmogorov we re-evaluated frozen checkpoints on frames drawn from *training* trajectories,

two raw frames from a training frame ($r = 0.97$ to the nearest one; raw-frame correlation $0.992 / 0.972 / 0.916$ at lags $1 / 2 / 4$): DMF-Gen $0.487 \rightarrow 0.483$, SiT $0.494 \rightarrow 0.491$, Senseiver $0.913 \rightarrow 0.873$, against an IDW control that moved $0.546 \rightarrow 0.537$ and therefore sets a $\approx 1.6\%$ frame-difficulty floor. Net of the controls, no method in either regime gains more than five percent on an observed channel from having seen its evaluation targets during training. At $\sim 6.5\text{M}$ parameters and 10^5 – 10^6 points per frame these models learn the sensing operator rather than a lookup table, and the protocol gap to the literature must come from elsewhere. (An earlier evaluation on a single snapshot of a 617-consecutive-frame cutout, made before the present protocol and its audit repairs, had suggested a factor of four between shuffled and disjoint splits; it is superseded by the measurement above.) The split remains a correctness requirement — a benchmark whose test frames are training frames measures nothing about generalization, and the artifact’s tooling enforces it — but its numerical effect at these data scales is small.

(ii) *What “1% sparsity” means.* Much of the visually striking literature conditions on uniformly downsampled coarse grids — every cell of a coarse lattice, i.e. band-limited full-field information with bounded gaps — or adds physics guidance (PDE-residual or forcing terms at sampling time^{8,10}). Our protocol uses randomly scattered points and purely data-driven conditioning: the same nominal percentage carries far less information and admits large sensor-free regions. Measured on Kolmogorov with the same checkpoint and frames, rearranging the 655 scattered sensors onto a uniform 25×26 lattice improves DMF-Gen from 0.487 to 0.367 relative L_2 (CRPS $0.235 \rightarrow 0.171$) and IDW from 0.546 to 0.452 — a 25% accuracy difference from sensor *layout* alone, before any change of model or split. The interpolation floor moves the same way because uniformly spaced sensors bound the worst-case distance to data, and reconstruction error at 1% sparsity is dominated by that distance: a benchmark that adopts uniform sensing because it is convenient reports a systematically easier problem, and no amount of split hygiene compensates. Combining the lattice *and* the seen-trajectory split gives 0.364, indistinguishable from the lattice alone, so at 2D data scales the protocol gap we can reproduce is sensor geometry rather than leakage. Our headline claim is consequently narrower than we first drew it: *protocol, not architecture* remains right, but within protocol it is the sensing layout that dominates, and the split matters mainly as a correctness requirement.

(iii) *What is shown.* Publication figures typically display ensemble means or favorable frames; we display single posterior samples on fixed, pre-registered frames (on the Kolmogorov gallery frame the four leading rows score 0.57–0.82 as single samples, against 0.42–0.50 for the same rows’ fifty-frame ensemble-mean averages), because single-sample behavior is what the physics metrics score. Ensemble size interacts with the same point: $K=8$ coverage understates calibration (Sec. II E), and ensemble means visually erase exactly the small scales the spectra measure.

(iv) *Matched budgets.* Every model trains at a matched $\sim 50\text{k}$ -step budget with disclosed tuning asymmetry, not to per-method convergence. Fairness costs polish; the cost table prices it.

The synthesis we intend a reader to take away: *apparent reconstruction quality in this literature is substantially a property of evaluation protocol, and only secondarily of architecture.* A benchmark that fixes the protocol first makes the remaining — genuine — architectural differences measurable.

VI. DISCUSSION

a. *Which method should a practitioner use?* The benchmark’s answer is conditional, and that is the point. If sensing is dense relative to the flow’s intrinsic dimension (laminar, periodic regimes) use gappy POD: on the cylinder it reaches 0.056 aggregate $\text{rel-}L_2$ against 0.093 for the best learned method, recovers the unobserved pressure channel, and costs a least-squares solve against a rank-20 basis (Sec. IV A). On observed channels of turbulence at $\sim 1\%$ density, training-free interpolation remains competitive with the best

learned methods — the case for learning begins where the problem is ill-posed: unobserved channels, sparser sensing, occluded sensor sets, and surface-confined sensing (Secs. IV B and V C, Table VIII). There, deterministic learners are structurally limited to conditional means; among generative routes, latent compression buys inference speed and global field structure at a measured dissipation-range floor in 3D, while ambient point-cloud generation buys 3D small-scale fidelity at higher inference cost — an ordering that inverts on Kolmogorov (Sec. IV C) — with geometry-freedom beyond a single body as the structural advantage the deferred regimes are designed to test. If the discretization exceeds $\sim 300^3$, grid families exit on measured walls; if it lives on a body-fitted or unstructured mesh, they exit on interface grounds, as the cylinder’s grid export already illustrates. Uncertainty from any current method must be recalibrated before operational use, and the benchmark ships the wrapper.

b. Limitations. All three regimes are simulation data on canonical flows, and all three are Cartesian or near-Cartesian — two periodic Cartesian grids and one body-fitted O-grid around a circular cylinder — so the present suite measures neither measurement realism nor geometric generality: the corrupted-operator study (Table VIII) uses synthetic noise and occlusion on one 2D regime, the surface-sensing study (Sec. IV B) a single body, and a multiphysics regime with realistic measurement operators (wildfire LES) and an unstructured, case-varying geometry regime with surface-only sensing (transonic wing family) are deferred to a follow-up paper, so claims about geometry-freedom beyond the cylinder are structural properties of the interfaces, not measured results. All learned rows train per-regime (no pretraining across flows); the suite contains no experimental data — particle-image-velocimetry or pressure-sensitive-paint regimes with real sensor noise are the natural next addition; single-snapshot reconstruction ignores temporal information that assimilation exploits; the cost accounting is single-GPU (H100) and does not probe multi-GPU scaling; the tuning-budget asymmetry of Sec. II F item (v) is disclosed but not eliminated; and training-seed variance is measured for the 2D fleets (Sec. IV C) but not for the 3D one, where a single replicate costs more than the rest of the suite combined.

VII. CONCLUSION

FlowRecon-Bench turns sparse-sensor flow reconstruction from a collection of incompatible per-paper evaluations into a measured landscape: three regimes spanning 2D laminar to 3D chaotic flow, eleven audited learned methods against four classical floors, proper probabilistic scoring, physics metrics with their estimator pitfalls documented, matched cost accounting, and leakage-controlled splits with the distortion of the dominant protocol quantified. The findings that outlive any single table: method ranking is regime-dependent and no method wins twice; the classical floor is sometimes unbeaten, and which classical method constitutes it changes with the regime; unobserved channels are near-unidentifiable in 3D turbulence yet recoverable on the cylinder by a linear basis, so identifiability is a property of the flow rather than of the method; mean-field error alone misranks generative methods, and on Kolmogorov it reverses the ranking between accuracy and CRPS across whole method families; nearly every scored ensemble is under-dispersed and post-hoc recalibration repairs stated coverage; and the computational axis separates grid from point families at measured walls. The datasets, splits, sensor manifests, configurations, audit notes, and evaluation driver are released for extension — the leaderboard is meant to be beaten.

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DATA AVAILABILITY

The benchmark artifact — datasets or regeneration recipes (JHTDB cutout coordinates and download scripts, OpenFOAM case definitions, converter scripts), split manifests, seeded sensor draws, per-method configurations, audit notes, and the evaluation driver — will be released on acceptance. [TODO: Zenodo/HF DOI; confirm JHTDB redistribution terms (fallback: recipe + coordinates); needs sign-off before anything goes public.]

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